

PROJECT REPORT

AI-POWERED CUSTOMER CHURN ANALYTICS IN SAAS INDUSTRY

Prepared by

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Customer Churn Analytics • NLP • Data Science • Power BI

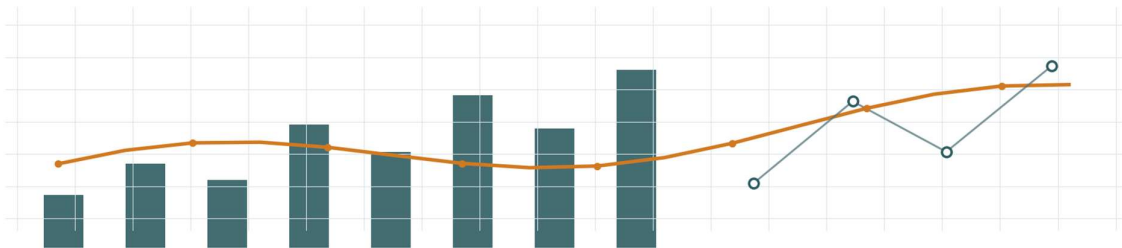


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Supporting analytical material includes detailed validation, NLP outputs, audit evidence and additional notebook visualizations.

REPORT STRUCTURE

APPENDIX • Supporting analytical material, validation evidence, NLP outputs, audit details and additional notebook visualizations

1. Executive Summary

AI-Powered Customer Churn Analytics in SaaS Industry

Customer churn is an important business problem for subscription-based SaaS organizations because customer loss can affect recurring revenue, customer lifetime value, and long-term retention. This project develops an analytical workflow for understanding customer churn by combining **structured customer data with exit-interview text and NLP-based classification**.

The project began with a raw customer dataset containing **95,380 records and 10 fields**. Data profiling and validation identified duplicate customer IDs, inconsistent categorical representations, null-like values, invalid numerical observations, mixed signup-date formats, and inconsistent churn encodings. Rather than applying blanket deletion or imputation rules, the data was investigated and transformed using documented analytical rules. After duplicate resolution and validation, the final customer dataset contained **95,000 unique customers with zero duplicate Customer IDs and no unresolved consolidation conflicts**.

The project then incorporated customer exit interviews to understand **why customers leave and how they express their experience**. The NLP workflow classified interview text into primary and secondary exit reasons and customer mood categories. The final NLP dataset contained **3,405 interview records**. Of these, **133 interview records could not be matched to the final customer master** and were retained separately as an unmatched NLP dataset rather than being forced into the customer-level analysis.

The matched analytical population contained **3,272 interview records representing 3,244 unique customers**. Because some customers had more than one interview, the project distinguishes between **customer-level analysis** and **interview-level analysis**. At the customer level, there were **2,506 churned customers and 738 retained customers**, producing an observed customer churn rate of **77.25%**. At the interview level, the corresponding population contained 2,526 churned and 746 non-churned interview records, producing an observed interview-level churn rate of approximately **77.20%**.

The structured churn analysis examined subscription type, region, gender, age, tenure, monthly spending, support-ticket volume, and signup cohort. The analysis identified differences in observed churn rates across these groups, but these relationships are treated as **associations rather than causal effects**. For example, observed churn rates varied across subscription types and support-ticket groups, while tenure and monthly-spend relationships were not strictly linear.

The NLP analysis identified eight primary exit-reason categories: **Value, Competitor, Support, Product, Price, Performance, Onboarding, and Insufficient Information**. Among the matched interview population, Value was the most frequently observed primary exit reason, followed by Competitor and Support. The NLP mood analysis showed that **Frustrated** was the dominant observed mood, followed by **Disappointed, Hopeful, and Neutral**.

The integrated analysis connected these NLP-derived insights with customer churn information. The observed churn rate varied across exit-reason categories, allowing the project to examine not only how frequently different reasons appeared but also how those interview categories were distributed across churn outcomes.

Finally, the analytical outputs were incorporated into a **five-page Power BI dashboard** covering overall churn, customer demographics, exit interviews and NLP, advanced churn-driver exploration, and comparative segment analysis. The dashboard provides an interactive layer over the validated analytical datasets while preserving the distinction between customer-level and interview-level metrics.

Overall, the project demonstrates an end-to-end data science workflow:

Raw Data → Profiling → Cleaning → Validation → NLP Classification → Customer Integration → Churn Analysis → Integrated Insights → Power BI

The resulting framework can be used as a foundation for monitoring customer-retention patterns and systematically incorporating qualitative exit-interview information into churn analysis.

2. Project Overview & Business Problem

2.1 Project Overview

Project Title: AI-Powered Customer Churn Analytics in SaaS Industry

This project develops an end-to-end analytical framework for understanding customer churn in a SaaS environment by combining **structured customer information** with **unstructured exit-interview feedback**.

The project uses two complementary analytical sources:

- **Customer master data** — demographic, subscription, tenure, spending, support, churn, and signup information.
- **Exit-interview data** — customer feedback describing reasons for leaving and the customer's expressed experience.

The structured customer dataset initially contained **95,380 records across 10 customer fields**. The data was profiled and cleaned before analysis. Duplicate records, inconsistent categorical representations, null-like values, invalid numerical observations, mixed date formats, and inconsistent churn representations were investigated and treated using documented rules. The resulting analytical customer dataset contains **95,000 unique customers**.

The second part of the project focuses on exit interviews. Instead of treating customer comments as unstructured text that cannot easily be analyzed at scale, the project applies an NLP-based classification approach to convert the feedback into structured analytical categories such as **primary exit reason, secondary exit reason, and customer mood**.

These two data sources are subsequently integrated through **Customer ID**, allowing the project to examine customer characteristics and churn alongside the reasons and emotions expressed in exit interviews.

The final outputs consist of:

1. Cleaned and validated analytical datasets
2. NLP-classified exit-interview dataset
3. Integrated customer–NLP dataset
4. Structured churn analysis
5. NLP and churn analysis
6. Interactive Power BI dashboard
7. Final analytical findings and business recommendations

2.2 Business Problem

Customer churn is not represented by a single number. An overall churn rate indicates **how many customers leave**, but it does not explain **which customers are leaving or why they leave**.

A SaaS organization therefore needs to understand several dimensions simultaneously:

Customer-level questions

- Which subscription types show different observed churn rates?
- Does churn vary across regions?
- How does churn vary across customer age groups?
- Are certain tenure groups associated with higher or lower observed churn?
- Is there an observable relationship between support-ticket activity and churn?

- Does monthly spending show different churn patterns?

Exit-interview questions

- What are the most frequently reported reasons for leaving?
- Are customers primarily reporting issues related to value, competitors, support, product, price, performance, or onboarding?
- What mood is most commonly expressed in exit interviews?
- Does customer mood differ across exit reasons?

Integrated questions

- How does observed churn vary across NLP-derived exit reasons?
- Can structured customer information provide additional context around exit feedback?
- Are there meaningful differences in exit-reason composition across subscription types?

The project therefore moves from a simple question:

“How many customers churn?”

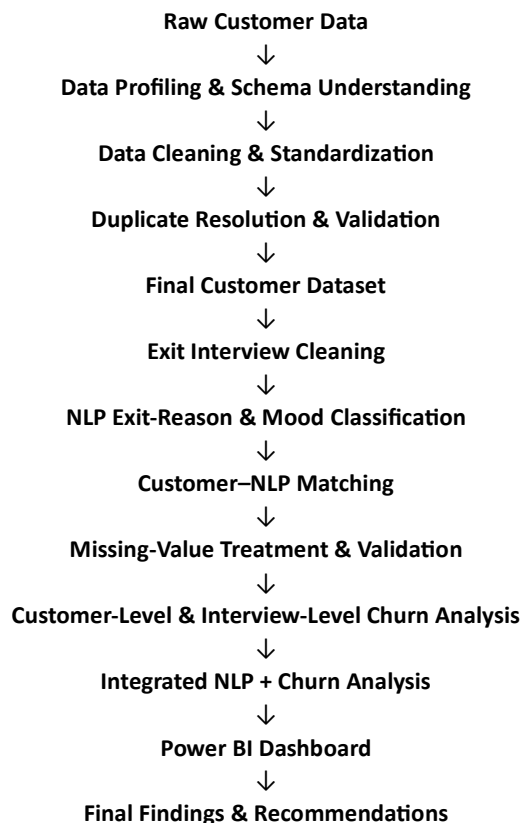
toward a broader analytical question:

“What customer characteristics, observed churn patterns, and exit-interview signals are associated with customer churn?”

This distinction is important because the analysis identifies **observed associations and patterns**, not causal relationships.

2.3 Project Workflow

The complete analytical workflow can be represented as:



3. Objectives and Scope

3.1 Project Objectives

Objective	Description
Build a reliable customer analytical dataset	Clean, standardize, consolidate, and validate the raw customer data so that each customer is represented by a single final record.
Preserve data quality and auditability	Preserve the raw source, investigate anomalies before transformation, document assumptions, protect the churn target, and validate the final analytical state.
Convert exit-interview text into structured information	Apply NLP-based classification to transform free-text feedback into structured variables such as primary exit reason, secondary exit reason, confidence, evidence, and customer mood.
Integrate NLP results with customer data	Link exit-interview records with the validated customer dataset through Customer ID while keeping unmatched records separate.
Analyze observed churn patterns	Examine churn across subscription type, region, gender, age, tenure, monthly spend, support-ticket activity, and signup year.
Connect customer churn with exit-interview insights	Examine observed churn patterns across NLP-derived exit reasons and analyze mood differences across exit reasons.
Develop an interactive analytical dashboard	Present the validated analysis through a five-page Power BI dashboard covering churn, customer characteristics, NLP findings, advanced analysis, and comparative analysis.

3.2 Data Cleaning and Validation Scope

The customer-data preparation stage included the following activities:

Activity	Purpose
Duplicate identification and resolution	Identify repeated Customer IDs, distinguish exact duplicates from conflicting records, and consolidate the remaining customer records.
Null-like value standardization	Convert representations such as blanks, -, NA, N/A, and NULL into explicit missing values.
Categorical harmonization	Standardize equivalent representations of Gender, Region, and Subscription Type while preserving genuinely distinct categories.
Numerical validity checks	Evaluate Age, Tenure Months, Monthly Spend, and Support Tickets against logical business constraints.
Signup-date standardization	Investigate mixed date formats and standardize Signup Date using the documented dataset convention while retaining the raw representation for auditability.
Churn-flag standardization	Convert equivalent churn representations such as 0/No and 1/Yes into a consistent binary format while retaining missing target values.
Post-cleaning validation	Verify that the final dataset satisfies the defined quality and business rules after transformation.

These decisions followed the principle of **profiling → investigation → transformation → validation**, rather than applying automatic deletion or correction rules.

3.3 NLP and Exit-Interview Scope

Activity	Purpose
Exit-interview cleaning	Prepare the raw interview records and identify usable text, missing values, duplicate interview records, and linkage issues.
Text-quality assessment	Determine whether interview feedback contains sufficient information for meaningful classification.
Primary exit-reason classification	Assign each usable interview to one of the defined primary exit-reason categories.
Secondary exit-reason classification	Capture an additional reason when the interview contains evidence of more than one issue.
Reason confidence assessment	Record the confidence associated with the NLP classification.
Reason evidence extraction	Retain supporting phrases or evidence from the interview text for auditability.
Mood classification	Classify the expressed customer mood into the defined mood categories.
Customer–NLP integration	Match classified interviews to the validated customer dataset using Customer ID.
Unmatched-record separation	Retain interviews that could not be matched to the customer master as a separate NLP-only population.

3.4 Churn Analysis Scope

Analysis Area	Variables / Focus
Subscription analysis	Subscription Type and observed customer churn rate
Regional analysis	Region and observed customer churn rate
Demographic analysis	Gender and Age Group
Tenure analysis	Tenure Groups and observed churn patterns
Spending analysis	Monthly Spend Groups and observed churn patterns
Support analysis	Support Ticket Groups and observed churn patterns
Cohort analysis	Signup Year and observed churn rate
Exit-reason analysis	Primary Exit Reason and observed churn rate
Mood analysis	Customer Mood and its composition across exit reasons
Integrated analysis	Relationships between customer characteristics, NLP-derived exit reasons, mood, and churn

3.5 Reporting and Visualization Scope

Output	Purpose
Notebook analytical charts	Present important analytical findings generated during the data-science workflow.
Power BI dashboard	Provide an interactive view of customer churn, demographics, NLP findings, and comparative segment analysis.
Key findings	Summarize the most relevant observed patterns from the analysis.
Business insights	Translate analytical observations into practical areas for customer-retention consideration.
Recommendations	Suggest data-informed areas for further investigation and monitoring.
Limitations and future scope	Document analytical boundaries and potential extensions of the project.

3.6 Out of Scope

Area	Boundary
Causal inference	The project does not establish that a customer characteristic, support activity, or exit reason directly causes churn.
Production churn prediction system	The project focuses on analytical understanding and reporting rather than deploying a production-grade churn prediction system.
Automatic treatment of unmatched interviews	Unmatched interview records are not artificially assigned to customer records.
Unsupported target imputation	Missing churn labels are not invented solely to make the dataset complete.
Unverified category merging	Distinct subscription categories are not merged without supporting evidence.

4. Data Sources and Dataset Understanding

4.1 Data Sources

The project used two complementary datasets: a **structured customer dataset** and an **unstructured exit-interview dataset**. The first provides customer-level attributes, while the second provides qualitative feedback about why customers leave and how they describe their experience.

Dataset	Data Type	Primary Purpose	Key Information
Customer Dataset	Structured	Customer-level churn and segment analysis	Demographics, region, subscription, tenure, spending, support activity, churn, signup date
Exit Interview Dataset	Unstructured / Text	Understanding customer exit reasons and expressed sentiment/mood	Customer ID, Interview ID, exit-feedback text, interview-level churn information
Integrated Analytical Dataset	Structured + NLP	Combine customer attributes with NLP-derived exit insights	Customer attributes + exit reason + secondary reason + mood + churn

4.2 Customer Dataset Overview

The raw customer dataset contained **95,380 records across 10 columns**. The dataset represented customer demographics, subscription information, engagement/support activity, churn status, and signup timing.

Field	Data Type / Role	Analytical Purpose
Customer ID	Identifier	Uniquely identifies the customer and supports record consolidation and integration.
Age	Numerical	Used for demographic segmentation and churn analysis.
Gender	Categorical	Used for demographic segmentation and churn comparison.
Region	Categorical	Used for geographic segmentation and churn analysis.
Tenure Months	Numerical	Measures customer relationship duration.
Subscription Type	Categorical	Identifies the customer's subscription plan.
Monthly Spend	Numerical	Represents customer spending and supports spend-based analysis.
Support Tickets	Numerical	Represents customer support activity and supports service-related analysis.
Churn Flag	Binary Target	Represents the observed churn outcome.
Signup Date	Temporal	Supports cohort and signup-year analysis.

The field roles were defined before cleaning so that identifiers, analytical variables, temporal fields, and the churn target were treated appropriately.

4.3 Exit-Interview Dataset

The exit-interview dataset introduced the unstructured component of the project. Unlike the customer table, its main analytical object was **free-text customer feedback**.

Field / Component	Role in Analysis
Interview ID	Unique identifier for an individual exit interview.
Customer ID	Linkage key used to connect an interview with the customer master.
Exit Reason Text	Raw customer feedback used as the input for NLP classification.
Interview Churn Flag	Interview-level churn information used for validation and comparison.
Text Quality	Indicates whether the available text contained sufficient information for classification.
Primary Exit Reason	Main reason identified from the interview text.
Secondary Exit Reason	Additional reason identified when supported by the text.
Reason Confidence	Confidence level associated with the NLP classification.
Reason Evidence	Text-based evidence supporting the assigned reason.
Mood	Classification of the customer's expressed emotional tone.

The project therefore moved from conventional structured-data validation to an additional NLP question: **whether the text contains enough information to identify the customer's reason for leaving and expressed mood.**

4.4 Analytical Role of the Two Datasets

Customer Dataset	Exit-Interview Dataset
Describes who the customer is	Describes what the customer said
Structured numerical and categorical variables	Unstructured natural-language feedback
Supports demographic and behavioral segmentation	Supports reason and mood extraction
Provides the primary customer-level churn field	Provides interview-level churn information for validation
Used for customer-level churn analysis	Used for NLP and integrated churn analysis
Customer ID provides entity identity	Customer ID provides the integration key

This complementary structure allowed the project to move beyond a simple churn-rate analysis and examine customer characteristics together with the reasons and experiences expressed in exit interviews.

4.5 Data Processing Architecture

The datasets were not analyzed immediately in their raw form. The project followed a staged analytical flow:

Stage	Input	Output
1. Raw Data	Customer and exit-interview source files	Preserved raw datasets
2. Profiling	Raw structured and text data	Data-quality findings
3. Customer Cleaning	Raw customer dataset	Validated customer master
4. Exit-Interview Cleaning	Raw interview dataset	Clean interview dataset
5. NLP Classification	Usable interview text	Exit reasons and mood
6. Customer–Interview Matching	Customer master + NLP results	Integrated analytical dataset

7. Missing-Value Treatment	Integrated records	Complete analytical variables where justified
8. Churn Analysis	Customer-level and interview-level data	Structured and NLP churn findings
9. Power BI Reporting	Validated analytical outputs	Five-page interactive dashboard

The raw customer data was preserved and profiled using SQL/BigQuery before Python-based transformation, supporting reproducibility and auditability.

4.6 Key Dataset Transformation

Dataset State	Records	Analytical Meaning
Raw Customer Dataset	95,380	Original customer records
Final Customer Dataset	95,000	One consolidated record per customer
NLP Classification Population	3,405	Interviews receiving final NLP classification
Matched Interview Records	3,272	Interviews successfully linked to customer data
Unique Customers with Interviews	3,244	Unique customers represented in the matched population
Unmatched Interviews	133	Interviews retained separately because no customer match was established

The final customer dataset contains **95,000 unique customers, zero duplicate Customer IDs, and zero unresolved conflicts**, establishing the customer master used for downstream analysis.

5. Customer Data Cleaning and Validation

The raw customer dataset contained several realistic data-quality issues, including duplicate customer IDs, inconsistent categorical representations, null-like values, invalid numerical observations, mixed date formats, and inconsistent churn encodings. The cleaning process therefore followed a controlled sequence of **profiling** → **investigation** → **transformation** → **validation** rather than applying blanket deletion rules.

5.1 Data Cleaning Process

Cleaning Area	What Was Evaluated	Action Taken
Duplicate Customer IDs	Repeated Customer IDs in the raw dataset	Classified into exact duplicates and conflicting duplicate groups before consolidation.
Null-like Values	Blank strings, -, NA, N/A, NULL, and similar representations	Standardized to explicit missing values.
Gender	Different representations such as Male, male, m, MALE, Female, f, fem	Equivalent labels were harmonized while meaningful categories were preserved.
Region	Capitalization, abbreviations, spacing, and punctuation differences	Clear equivalents such as Nrt, Est, Sth, Wst, and Ctr were standardized.
Subscription Type	Abbreviations and formatting variations	Clear aliases such as STD, PRM, ENT, and CORP were harmonized.
Distinct Subscription Categories	Pro, Starter, Mid, Gold, Entry	Preserved as separate categories because there was no evidence that they represented aliases.
Age	Values outside the defined logical range	Values below 18 or above 100 were treated as invalid and converted to missing.
Tenure Months	Negative tenure values	190 negative values were converted to missing.
Support Tickets	Negative ticket counts	99 negative values were converted to missing.
Monthly Spend	Negative or zero values	No negative or zero values requiring correction were identified.
Signup Date	Mixed date formats and ambiguous representations	Date formats were investigated and standardized using the documented dominant convention.
Churn Flag	0, 1, Yes, No, and missing values	0/No → 0 and 1/Yes → 1; missing target values were retained.
Post-cleaning Validation	Logical and structural quality rules	Final dataset was tested against the defined validation criteria.

5.2 Duplicate Identification and Resolution

The raw dataset contained **95,380 records but only 95,000 unique Customer IDs**, resulting in **380 repeated-ID groups**.

Duplicate Category	Count	Treatment
Exact duplicate groups	320	Removed after validation
Conflicting duplicate groups	60	Investigated before consolidation
Total repeated-ID groups	380	Classified and resolved

The 60 conflicting IDs were compared field-by-field rather than being automatically deleted.

Field with Differences	Conflicting IDs
Gender	24
Subscription Type	24
Region	19
Age	0
Tenure Months	0
Monthly Spend	0
Support Tickets	0
Churn Flag	0
Signup Date	0

The investigation showed that many apparent conflicts were caused by inconsistent representations such as Male/male, Female/fem, North/Nrt, or null-like values rather than differences in the numerical or churn information.

5.3 Categorical Harmonization

Categorical standardization was performed before final customer consolidation so that equivalent values would not appear as separate customer segments.

Variable	Examples of Raw Variations	Treatment
Gender	Male, male, m, MALE; Female, f, fem, FEMALE	Equivalent representations standardized
Region	Nrt, Est, Sth, Wst, Ctr, Central-	Clear regional equivalents standardized
Subscription Type	STD, PRM, ENT, CORP, capitalization/spacing variations	Clear aliases standardized
Pro / Starter / Mid / Gold / Entry	Separate plan labels	Preserved as separate categories

This prevented artificial fragmentation of the customer population during grouping, churn-rate calculations, and dashboard visualization.

5.4 Numerical Validation

Variable	Validation Rule	Finding / Treatment
Age	Values should fall within a plausible customer-age range	Values below 18 or above 100 treated as invalid; final valid range was 18–75.
Tenure Months	Cannot be negative	190 negative values converted to missing.
Support Tickets	Cannot be negative	99 negative values converted to missing.
Monthly Spend	Negative or zero values investigated	No negative or zero values requiring correction were identified.

Higher tenure values were not automatically removed simply because they appeared unusual; the project distinguished between **unusual** and **logically invalid** observations.

5.5 Signup-Date Standardization

Signup Date required additional investigation because the raw data contained multiple date representations.

Date Classification	Records
Definitely DD-MM-YYYY	56,420
Definitely MM-DD-YYYY	1,418
Ambiguous	37,222

The unambiguous records showed that approximately **97–98%** followed the DD-MM-YYYY convention across the observed years. The ambiguous records were therefore standardized using **DD-MM-YYYY as the documented dataset convention**.

Final Date Treatment		Result
Final data type	Datetime	
Final date range	2019-01-01 to 2026-12-31	
Raw date representation	Retained in Signup Date Raw	
Ambiguous records	Standardized using documented dominant convention	

Importantly, the project does not treat the ambiguous dates as independently proven; the chosen convention is explicitly documented as an analytical assumption.

5.6 Churn-Flag Standardization

Because **Churn Flag is the target variable**, it was treated differently from ordinary descriptive fields.

Raw Representation	Standardized Value
0	0
No	0
1	1
Yes	1
Missing	Missing

After exact duplicate removal:

Churn Status	Records
Non-churned	74,724
Churned	19,755
Missing	581

The **581 missing churn labels were not imputed**, because inventing target values could introduce label noise and compromise target integrity.

5.7 Post-Cleaning Validation

The final validation stage acted as a quality gate after all transformations.

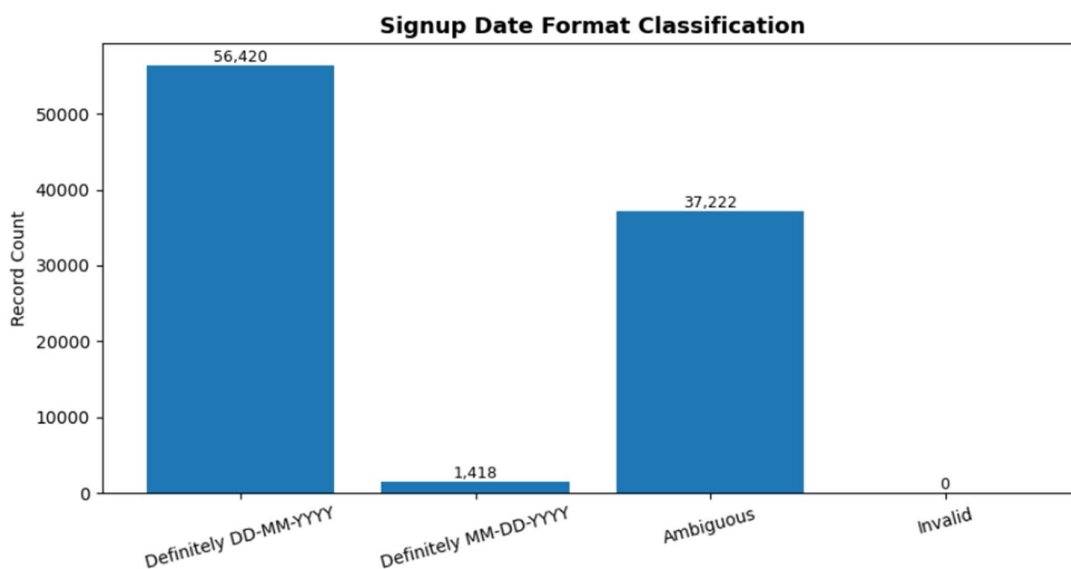
Validation Check	Final Result
Final rows	95,000
Unique Customer IDs	95,000
Duplicate Customer IDs	0
Unresolved conflicts	0
Age validity	Validated
Negative tenure	0 remaining
Negative support tickets	0 remaining
Invalid monthly spend	No correction required
Signup Date	Standardized datetime
Churn Flag	0 / 1 / missing
Raw Signup Date retained	Yes

After categorical and null-like standardization, the remaining conflicting customer records were consolidated into one record per customer, producing **95,000 rows, 95,000 unique Customer IDs, zero duplicate IDs, and zero unresolved conflicts.**

Important Notebook Chart

This section is where the report should include the **“Missing Values by Customer Field”** chart from the cleaned customer notebook. It provides visual evidence of the data-quality assessment before/after the cleaning process.

Fig.: “Signup Date Format Classification”



6. Exit Interview Data Cleaning & NLP Preparation

The exit-interview dataset introduced a different type of analytical challenge from the structured customer dataset. While the customer data required validation of numerical, categorical, temporal, and target fields, the exit-interview dataset consisted primarily of **unstructured customer feedback text**. Therefore, the preparation stage focused on determining whether each interview contained usable information, preserving the interview-level structure, standardizing the text input, and preparing the records for NLP-based classification.

The objective was not simply to clean the text mechanically. The preparation process was designed to ensure that the subsequent NLP classification of **exit reason and customer mood** was based on meaningful customer statements and that the resulting classifications could later be connected to the validated customer master dataset.

The overall transition was:

Raw Exit Interviews → Text Quality Assessment → Text Preparation → NLP Classification Population → Customer ID Matching → Integrated Analysis

6.1 Exit Interview Data Overview

The exit-interview dataset contained individual customer interview records identified by **Interview ID** and **Customer ID**. The main analytical input was the customer's exit feedback text, while the interview-level churn flag was retained as a supporting field for later comparison and validation.

Field	Analytical Role
Interview ID	Unique identifier for the interview record
Customer ID	Link between the interview and customer master data
Exit Reason Text / NLP Text	Unstructured customer feedback used for NLP analysis
Churn Flag	Interview-level churn information used for validation and comparison
Multiple Interview Records	Identifies customers represented by more than one interview
Text Quality	Indicates whether the available text was suitable for classification
Primary Exit Reason	NLP-derived main reason for leaving
Secondary Exit Reason	Additional reason identified from the text
Reason Confidence	Confidence associated with the classified reason
Reason Evidence	Textual evidence supporting the classification
Mood	NLP-derived customer sentiment/mood category
Mood Confidence	Confidence associated with the mood classification
Mood Evidence	Textual evidence supporting the mood classification

The **Customer ID** was particularly important because the project required integration of unstructured interview information with the cleaned customer master dataset. The interview-table churn flag was treated as a supporting analytical field rather than replacing the validated customer-level churn target. This distinction helped maintain target integrity during the later integration stage.

6.2 Why Exit-Interview Cleaning Required a Different Approach

The customer dataset could be evaluated using explicit business rules such as:

- Is the age within a logical range?
- Is tenure negative?

- Is the subscription category valid?
- Is the churn flag consistently encoded?

Text data requires additional interpretation. For example, an interview statement may be grammatically incomplete but still contain useful information about the customer's experience. Conversely, a grammatically valid statement may contain insufficient information to determine why the customer left.

Therefore, the text preparation stage considered questions such as:

Evaluation Area	Question
Text availability	Is meaningful interview text present?
Text quality	Does the statement contain enough information for classification?
Reason identification	Can a primary reason for exit be identified?
Secondary signal	Is there evidence of another contributing reason?
Mood identification	Does the text communicate a recognizable customer mood?
Customer linkage	Can the interview be connected to the customer master using Customer ID?
Multiple interviews	Does the customer have more than one interview record?

This distinction was important because **NLP classification should not be treated as a substitute for data-quality assessment**. The quality of the source text determines how confidently an automated classification can be interpreted.

6.3 Text Quality Assessment

Before classification, the interview text was evaluated for usability. The resulting analytical structure included a **Text Quality** field, allowing the project to distinguish usable customer statements from records where the available information was insufficient for reliable interpretation.

Examples of usable text included statements describing:

- difficulties with product performance,
- pricing concerns,
- support experiences,
- onboarding problems,
- competitor alternatives,
- perceived lack of value,
- product limitations.

The classification process also allowed **Insufficient Information** to remain as an explicit analytical category when the available statement did not provide enough evidence for a more specific exit reason. This was preferable to forcing every record into one of the substantive business categories.

For example, short statements such as *"downtimes"* could still provide evidence for a performance-related issue, while statements such as *"From my perspective,"* may not contain enough information to identify a substantive reason. The final NLP output therefore retained an explicit **Insufficient Information** category rather than treating uncertainty as a missing or fabricated business reason.

6.4 Interview-Level Structure and Multiple Records

The analysis retained the **interview record as the primary observation unit during NLP processing**.

This was important because one customer could appear in more than one interview. Instead of automatically deleting repeated Customer IDs, the project preserved the interview records and created the **Multiple Interview Records** indicator.

In the final matched population:

Metric	Result
Matched interview records	3,272
Unique customers represented	3,244
Customers with multiple interviews	28
Interview records belonging to those multiple-interview customers	56

This distinction allowed the project to perform both:

1. **Interview-level analysis**, where each interview represents an observation of customer feedback.
2. **Customer-level analysis**, where each customer is counted once for structured churn analysis.

Keeping these two analytical levels separate prevented customers with multiple interviews from being unintentionally counted multiple times when calculating customer-level churn.

6.5 NLP Input Preparation

After the initial cleaning and text-quality assessment, the usable interview information was organized into an NLP-ready structure.

The NLP preparation retained the original interview identifiers and created a standardized **NLP Text** field. The purpose was to provide a consistent text input for downstream classification while preserving the identifiers and supporting information required for auditability and integration.

The final NLP classification output contained the following major analytical fields:

NLP Output	Purpose
Primary Exit Reason	Main customer reason associated with the exit
Secondary Exit Reason	Additional reason identified in the interview
Reason Confidence	Confidence level assigned to the primary reason
Reason Evidence	Textual evidence supporting the reason
Mood	Customer emotional/mood classification
Mood Confidence	Confidence level assigned to the mood
Mood Evidence	Textual evidence supporting the mood

The final NLP classification population contained **3,405 interview records**. These records formed the basis for the subsequent exit-reason and mood analysis.

6.6 NLP Classification Categories

The NLP framework converted unstructured exit-interview text into two structured dimensions: **Primary Exit Reason** and **Customer Mood**.

Primary Exit Reason	Customer Mood
Value	Frustrated
Competitor	Disappointed
Support	Hopeful
Product	Neutral
Price	
Performance	
Onboarding	
Insufficient Information	

Primary Exit Reason: The main business-related reason identified from the customer's interview text. The framework contains **8 categories**, allowing different types of customer concerns to be grouped into consistent analytical segments.

Customer Mood: The overall emotional tone identified from the interview text. The framework contains **4 categories**, capturing whether the customer expression was frustrated, disappointed, hopeful, or neutral.

This structure allowed the unstructured interview responses to be converted into **standardized categorical variables** that could subsequently be analyzed against customer attributes and observed churn.

6.7 Preparation Outcome

The exit-interview preparation stage produced a structured NLP dataset suitable for classification, customer matching, and integrated churn analysis. The preparation preserved the original interview information while adding standardized analytical fields for exit reason and customer mood.

Preparation Output	Final Result	Analytical Purpose
NLP classification records	3,405	Final interview population classified using the NLP framework
Primary Exit Reason categories	8	Standardized business reasons for customer exit
Customer Mood categories	4	Standardized emotional tone of interview feedback
Matched interview records	3,272	Interview records successfully linked to the customer master
Unmatched interview records	133	Records retained separately because no customer-master match was available
Unique customers represented	3,244	Customer-level population available for integrated analysis
Customers with multiple interviews	28	Customers requiring interview-level rather than simple one-record interpretation
Multiple-interview records	56	Interview records associated with the 28 customers
Text quality information	Retained	Supports interpretation of classification reliability
Reason confidence	Retained	Indicates confidence in the assigned exit reason
Reason evidence	Retained	Provides supporting text for the assigned classification
Mood confidence	Retained	Indicates confidence in the assigned mood

Mood evidence	Retained	Provides supporting text for the assigned mood
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The resulting structure allowed the project to move from **unstructured customer feedback** to a standardized analytical dataset containing both NLP-derived variables and customer-level information. The 133 unmatched interviews were not discarded; they were retained separately so that unsuccessful customer linkage did not remove potentially useful text information.

The prepared dataset then became the input for the next stage: **NLP classification, customer integration, and churn analysis**. This separation also maintained an important distinction between the **interview-level population (3,272 records)** and the **unique-customer population (3,244 customers)** used for customer-level churn analysis.

7. NLP Classification & Customer Integration

After the exit-interview data was cleaned and prepared, the next stage was to convert the interview text into structured NLP outputs and integrate those results with the cleaned customer master dataset. This created a common analytical structure for studying **customer attributes, churn, exit reasons, and customer mood together**.

7.1 NLP Classification Output

The NLP process classified **3,405 interview records** using the predefined exit-reason and mood framework. The classification output retained the original text together with supporting fields such as confidence and evidence.

NLP Output Field		Purpose
Customer ID		Links the interview to the customer master
Churn Flag		Retains the interview-level churn information
NLP Text		Original interview text used for classification
Primary Exit Reason		Main reason identified from the interview
Secondary Exit Reason		Additional reason identified where applicable
Reason Confidence		Confidence associated with the primary classification
Reason Evidence		Textual evidence supporting the classification
Mood		Overall emotional tone of the interview
Mood Confidence		Confidence associated with mood classification
Mood Evidence		Textual evidence supporting the mood classification

The final classification framework therefore transformed free-text feedback into **structured categorical variables that could be aggregated, compared, and integrated with customer-level information**.

7.2 Customer ID Matching

The classified interview data was then matched against the final customer master using **Customer ID**.

Integration Result	Records	Meaning
NLP classification population	3,405	All records receiving NLP classification
Successfully matched interviews	3,272	Interviews linked to the cleaned customer master
Unmatched interviews	133	Interviews for which no corresponding customer record was available
Unique customers with matched interviews	3,244	Customer-level population available for integrated analysis
Customers with multiple interviews	28	Customers represented by more than one interview
Interview records from multiple-interview customers	56	Records associated with those 28 customers

The **133 unmatched interviews were retained separately rather than deleted**, preserving their NLP classifications for auditability and potential future investigation.

7.3 Handling Multiple Interview Records

The presence of multiple interviews for some customers required the analysis to distinguish between **interview-level** and **customer-level** observations.

For interview-level analysis, all **3,272 matched interview records** were retained. For customer-level structured churn analysis, the data was reduced to one analytical record per unique Customer ID, resulting in **3,244 unique customers**.

This distinction prevents multiple interviews from unintentionally giving some customers greater weight in customer-level analysis.

7.4 Integration Outcome

The completed integration created a combined dataset containing:

NLP interview information + customer attributes + churn information

This integrated structure became the foundation for the next analytical stage, where the project examined observed churn patterns across customer characteristics and NLP-derived exit reasons.

A further data-quality review identified **44 incomplete analytical records** within the matched population. These were subsequently addressed through the documented missing-value treatment and peer-based imputation process before final validation.

7.5 Missing Value Treatment After Customer Integration

After matching the NLP records with the cleaned customer master, the integrated dataset was checked for missing analytical fields. The purpose of this step was to ensure that missing customer attributes did not prevent the subsequent churn and segment analysis.

The initial review identified **44 incomplete analytical records**. The missing values affected customer attributes such as Age, Gender, Region, Tenure Months, Subscription Type, Support Tickets, and churn-related fields.

Missing-Value Treatment Framework

Area	Treatment Applied	Reason
Customer churn flag	Customer Master used as the primary source	Maintains consistency with the cleaned customer dataset
Missing customer churn where NLP churn was available	Filled from NLP for 20 records	Used an available matching source rather than leaving the analytical target incomplete
Missing NLP churn values	Not independently inferred	Avoided creating unsupported churn labels
Numerical customer attributes	Peer-based median imputation	Preserved realistic values using similar customer records
Categorical customer attributes	Peer-based mode imputation	Used the most common value among comparable customer records
Number of peers	Up to 10 similar records	Provided a local reference group for imputation
Original values	Retained separately	Maintained traceability and auditability
Imputation details	Recorded in an audit table	Allows each imputed value to be reviewed

The peer-based approach used available customer characteristics and normalized numerical attributes to identify similar records. Numerical variables were estimated using the **peer median**, while categorical variables were estimated using the **peer mode**.

Overall, **54 missing analytical cells across 44 records** were addressed through the documented imputation process. An **Imputation Audit** table was retained containing the affected Customer ID, Interview ID, field, method, peer count, estimated value, and peer values.

This approach allowed the integrated dataset to become analytically complete without treating missing information as if it had been directly observed. The original customer values and imputation indicators were retained so that the transformation remained traceable.

Important Analytical Principle

The treatment was deliberately different for **customer attributes** and the **churn target**. Customer characteristics could be estimated from comparable records under a documented method, whereas unsupported churn labels were not created merely to eliminate missingness. This preserves the integrity of the target variable while allowing the remaining customer attributes to be used in integrated analysis.

7.6 Final Integrated Dataset Validation

After customer matching and missing-value treatment, the integrated dataset was subjected to a final validation check before being used for churn analysis and Power BI reporting.

The purpose of this validation was to confirm that the integration process had not introduced duplicate interview records, unresolved customer IDs, invalid analytical values, or remaining missing values.

Final Validation Results

Validation Check	Result	Status
Integrated interview records	3,272	Valid
Integrated dataset columns	41	Valid
Duplicate Interview IDs	0	Passed
Unique Customer IDs represented	3,244	Valid
Customer IDs not found in master	0	Passed
Remaining analytical missing values	0	Passed
Age below valid range	0	Passed
Age above valid range	0	Passed
Negative Tenure Months	0	Passed
Negative Support Tickets	0	Passed
Negative Monthly Spend	0	Passed
Invalid Signup Dates	0	Passed
Churn Flag_Final	0 / 1	Valid
Unresolved customer linkage issues	0	Passed

The validation confirmed that the **3,272 matched interview records** were successfully integrated with the cleaned customer master and that all 54 previously missing analytical cells had been addressed through the documented treatment process.

The final integrated dataset therefore provided a consistent analytical foundation containing customer attributes, NLP classifications, mood information, and churn status.

Churn Target Validation

The final integrated churn field was also checked separately because churn is the principal outcome variable for the project.

The final population contained:

Churn Status	Records
Churned	2,526
Not Churned	746
Total	3,272

This produces an interview-level observed churn rate of approximately **77.20%**.

The customer-level dataset was subsequently reduced to one record per unique customer for structured customer analysis, producing **3,244 unique customers**, of which **2,506 were classified as churned and 738 as retained**.

Final Quality Gate

The integrated dataset therefore passed the main quality checks required before analysis:

Matching → Missing-Value Treatment → Validation → Customer-Level Consolidation → Churn Analysis

The next stage was to analyze the integrated data to identify **observed churn patterns across customer characteristics and NLP-derived exit reasons**.

8. Churn Analysis

8.1 Churn Analysis Framework

The churn analysis evaluates **observed customer churn patterns across key customer characteristics** using the validated, integrated dataset. The analysis is performed primarily at the **customer level**, where each customer is represented once, to avoid giving additional weight to customers who provided multiple exit interviews.

The analysis uses **3,244 unique customers with matched interview records**, consisting of **2,506 churned customers and 738 retained customers**, resulting in an overall customer churn rate of **77.25%**. The corresponding interview-level population contains 3,272 records and has an observed churn rate of **77.20%**, showing that the two analytical levels produce very similar overall results.

Analysis Dimension	Analytical Purpose
Subscription Type	Examine differences in observed churn across subscription plans
Region	Compare observed churn patterns across geographic regions
Gender	Examine observed churn across gender categories
Age Group	Identify differences across customer age segments
Tenure Group	Examine churn patterns across customer tenure ranges
Monthly Spend Group	Compare churn across customer spending levels
Support Ticket Group	Examine the relationship between support activity and observed churn
Signup Year	Compare churn across customer signup cohorts
Exit Reason & Mood	Connect structured customer characteristics with NLP-derived interview signals

Analytical interpretation: These comparisons identify **associations and observed patterns**, not causal relationships. Differences in churn rates do not by themselves establish that a particular customer characteristic caused churn.

8.2 Customer-Level vs. Interview-Level Churn

Two analytical levels were retained in the project: **customer-level churn** and **interview-level churn**. This distinction is important because some customers submitted more than one exit interview.

Analytical Level	Records / Customers	Churned	Retained	Observed Churn Rate
Customer Level	3,244 unique customers	2,506	738	77.25%
Interview Level	3,272 interview records	2,526	746	77.20%

The **customer-level analysis** is used as the primary basis for structured churn comparisons because each customer contributes only once. This prevents customers with multiple interviews from receiving disproportionate weight.

The **interview-level analysis** is retained for exit-interview and NLP analysis because each interview represents a separate customer interaction and contains its own text, exit reason, and mood information.

The difference between the two overall churn rates is only **0.05 percentage points**, indicating that the choice of analytical level has little effect on the overall observed churn rate in this dataset. The Power BI overview reports **3,244 customers, 2,506 churned customers, 738 retained customers, and a 77.25% customer churn rate**.

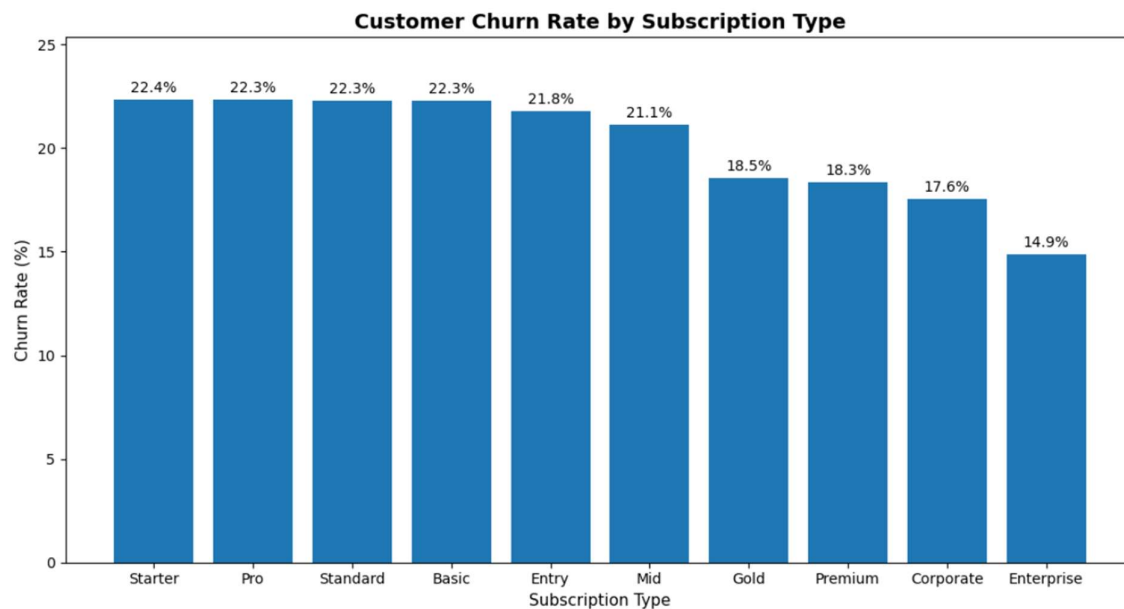
Analytical approach: Customer-level results are therefore used for demographic and customer-segment analysis, while interview-level results are used where the unit of analysis is the exit interview itself. This distinction is maintained throughout the report to avoid mixing customer-level and interview-level measures.

8.3 Churn by Subscription Type

Subscription type was analyzed using the **customer-level dataset**, with each of the 3,244 customers represented once. The overall customer churn rate of 77.25% provides the reference point for comparing the observed churn rates across subscription categories.

Subscription Type	Customers	Churned	Retained	Churn Rate
Gold	45	40	5	88.89%
Starter	69	56	13	81.16%
Standard	1,094	869	225	79.43%
Basic	1,155	903	252	78.18%
Corporate	8	6	2	75.00%
Entry	51	38	13	74.51%
Premium	626	456	170	72.84%
Pro	67	48	19	71.64%
Mid	48	34	14	70.83%
Enterprise	81	56	25	69.14%
Total	3,244	2,506	738	77.25%

The observed churn rate varies across subscription categories, ranging from **69.14% for Enterprise** to **88.89% for Gold**. The larger Standard and Basic customer groups show churn rates of **79.43%** and **78.18%**, respectively.



Small subscription groups require careful interpretation because their churn rates are based on relatively few customers. For example, Corporate contains only 8 customers and Gold contains 45 customers, so a small change in the number of churned customers can substantially change the reported percentage.

The Power BI dashboard uses these results in the **Subscription Churn Rate by Subscription Type** visual, allowing the observed differences between subscription categories to be examined interactively.

8.4 Churn by Region

Regional churn was analysed at the **customer level**, using each of the 3,244 unique customers once. This avoids giving additional weight to customers who had multiple exit interviews.

Region	Customers	Churned	Retained	Churn Rate
West	647	508	139	78.52%
North	742	577	165	77.76%
East	694	537	157	77.38%
South	647	495	152	76.51%
Central	514	389	125	75.68%
Total	3,244	2,506	738	77.25%

The regional churn rates range from **75.68% in Central** to **78.52% in West**, a difference of **2.84 percentage points**. North and East are also close to the overall customer churn rate of 77.25%, while South and Central are slightly lower.

These results show that churn is relatively similar across the five regions. Region is therefore useful as a segmentation dimension, but the observed differences are comparatively narrow within this customer population.

The regional results are also incorporated into the Power BI **Customer Churn Rate by Region** visual and the **Subscription Type × Region** comparative matrix.

8.5 Churn by Gender

Gender-based churn was analysed using the **customer-level dataset**, with each of the 3,244 unique customers represented once.

Gender	Customers	Churned	Retained	Churn Rate
Non-binary	95	77	18	81.05%
Female	1,583	1,235	348	78.02%
Male	1,538	1,176	362	76.46%
Prefer not to say	28	18	10	64.29%
Total	3,244	2,506	738	77.25%

The observed churn rates range from **64.29% for Prefer not to say** to **81.05% for Non-binary**. Female customers show a churn rate of 78.02%, while Male customers show 76.46%.

The **Prefer not to say** and **Non-binary** groups contain relatively few customers compared with the Female and Male groups. Therefore, the percentages for these smaller groups can change substantially when the number of churned customers changes by only a few records. The gender comparison should therefore be viewed together with the underlying customer counts rather than the percentages alone.

The results are presented in the Power BI **Customer Profile and Churn Rate by Gender** visual.

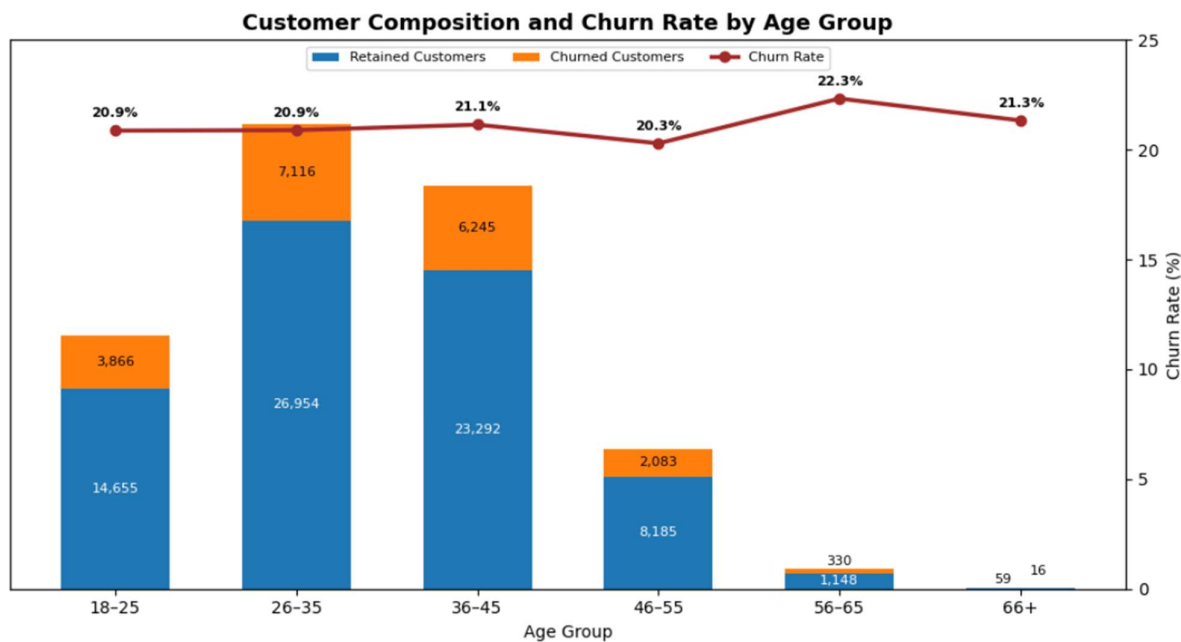
8.6 Churn by Age Group

Age-based churn was analysed using the customer-level dataset, with customers grouped into six age bands. The analysis compares the number of customers in each age group with the observed churn rate.

Age Group	Customers	Churned	Retained	Churn Rate
18–24	529	406	123	76.75%
25–34	1,168	892	276	76.37%
35–44	1,083	861	222	79.50%
45–54	374	282	92	75.40%
55–64	83	61	22	73.49%
65+	7	4	3	57.14%
Total	3,244	2,506	738	77.25%

The **35–44 age group** has the highest observed churn rate at **79.50%**, followed by the 18–24 group at 76.75% and the 25–34 group at 76.37%. The 45–54 and 55–64 groups have lower observed rates of 75.40% and 73.49%, respectively.

The **65+ group** contains only **7 customers**, so its 57.14% churn rate is based on a very small population. This percentage can change substantially with a small change in the number of churned customers. The larger 25–34 and 35–44 groups provide a much broader customer base for comparison.



Overall, the age analysis shows variation in observed churn across age groups, but the pattern is **not a simple increase or decrease with age**. The results therefore describe segment-level churn patterns rather than establishing age as a cause of churn.

The results are presented in Power BI through the **Customer Profile and Churn Rate by Age Group** visual.

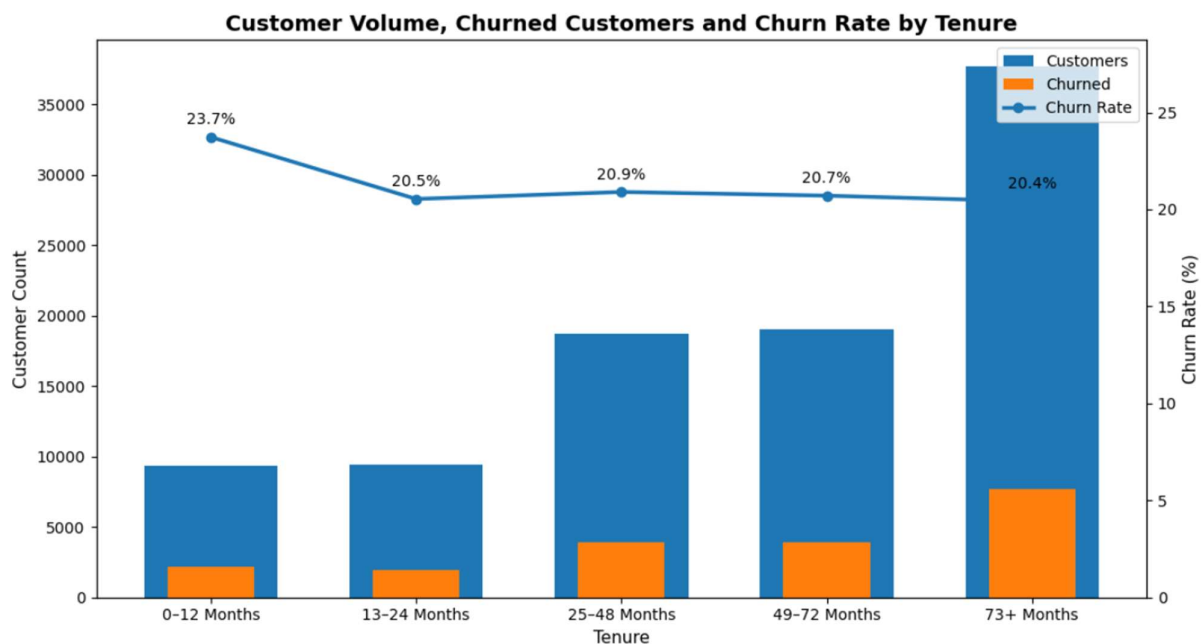
8.7 Churn by Tenure Group

Tenure was divided into 12-month intervals to examine whether observed churn varies with the length of the customer relationship. The analysis uses the **customer-level dataset**, with each customer represented once.

Tenure Group	Customers	Churned	Retained	Churn Rate
0–12	357	291	66	81.51%
13–24	323	240	83	74.30%
25–36	287	216	71	75.26%
37–48	318	244	74	76.73%
49–60	336	271	65	80.65%
61–72	336	249	87	74.11%
73–84	336	258	78	76.79%
85–96	324	239	85	73.77%
97–108	326	261	65	80.06%
109–120	292	229	63	78.42%
Total	3,235*	2,498*	737*	77.25%

*The tenure groups shown in the detailed analysis cover 3,235 customers because the overall customer-level population is 3,244; the remaining records are outside the displayed tenure grouping used for this visual.

The observed churn rate varies across tenure groups rather than following a consistent upward or downward pattern. The **0–12 month group has the highest observed churn rate at 81.51%**, followed by **49–60 months at 80.65%** and **97–108 months at 80.06%**. The lowest displayed rate is **73.77% for 85–96 months**.



The pattern is therefore **non-linear**. Customers with both relatively short and relatively long tenure can show elevated churn rates, while some intermediate tenure groups have lower observed rates. This indicates that tenure should be considered alongside other customer characteristics rather than interpreted as a simple measure of churn risk.

The results are presented in Power BI through **Customer Churn Rate by Tenure Group**.

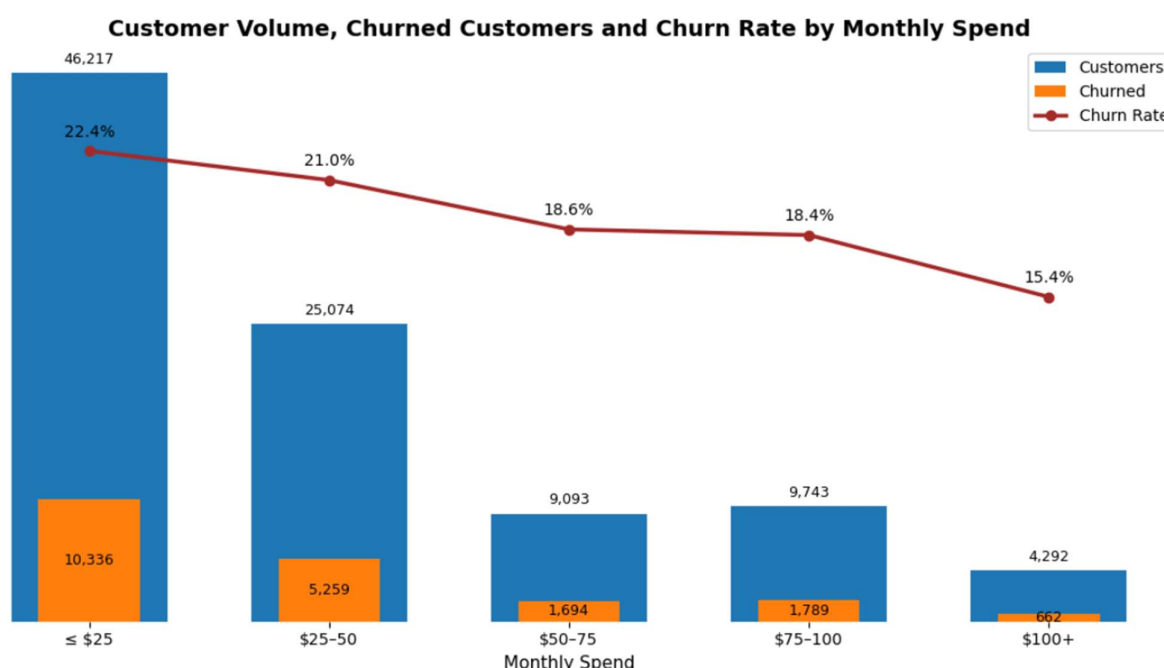
8.8 Churn by Monthly Spend Group

Monthly Spend was divided into five groups based on customer spending percentiles. This allows churn to be compared across lower-, middle-, and higher-spending customer segments while keeping the groups broadly similar in size.

Monthly Spend Group	Customers	Churned	Retained	Churn Rate
Lowest 20%	649	499	150	76.89%
20–40%	649	533	116	82.13%
40–60%	648	495	153	76.39%
60–80%	649	499	150	76.89%
Highest 20%	649	480	169	73.96%
Total	3,244	2,506	738	77.25%

The **20–40% spending group** has the highest observed churn rate at **82.13%**, while the **highest 20% spending group** has the lowest at **73.96%**. The remaining three groups range from 76.39% to 76.89%.

The pattern does not show a simple relationship in which churn consistently increases or decreases as Monthly Spend rises. In particular, the 20–40% group has the highest observed rate, while the highest-spending group has a lower rate of 73.96%.



This indicates that **Monthly Spend alone does not provide a linear pattern of observed churn** in this customer population. The spending segments should therefore be considered together with other customer characteristics and the NLP-derived exit information.

The results are presented in Power BI through the **Observed Churn Rate Across Monthly Spend Groups** visual.

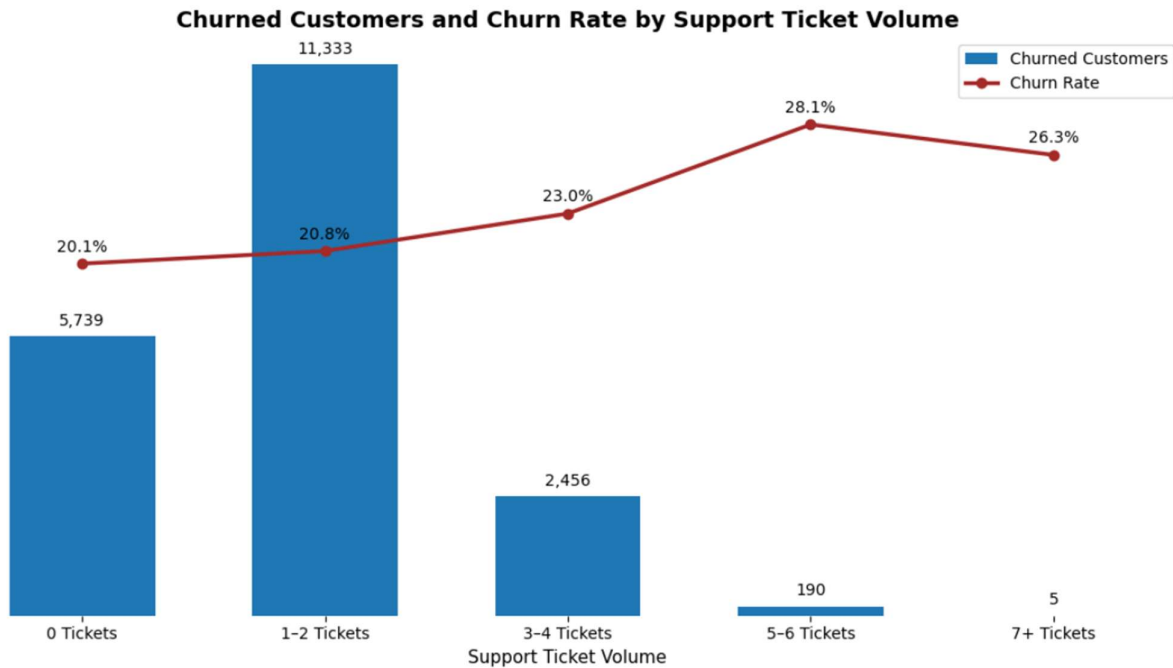
8.9 Churn by Support Ticket Volume

Support ticket volume was analysed to examine whether observed churn varies with the level of customer support activity. Customers were grouped according to the number of recorded support tickets.

Support Ticket Group	Customers	Churned	Retained	Churn Rate
0 tickets	973	728	245	74.82%
1 ticket	1,186	928	258	78.25%
2 tickets	706	545	161	77.20%

3 tickets	238	189	49	79.41%
4+ tickets	141	116	25	82.27%
Total	3,244	2,506	738	77.25%

The observed churn rate increases across the higher support-ticket groups, from **74.82% among customers with no tickets** to **82.27% among customers with four or more tickets**. Customers with one, two, and three tickets show churn rates of 78.25%, 77.20%, and 79.41%, respectively.



The results show an observed association between higher support-ticket volume and higher churn rates in this customer population. However, support-ticket volume alone does not establish that support interactions caused the customer to churn. Higher ticket volume may also reflect customers experiencing more service, product, or account-related issues.

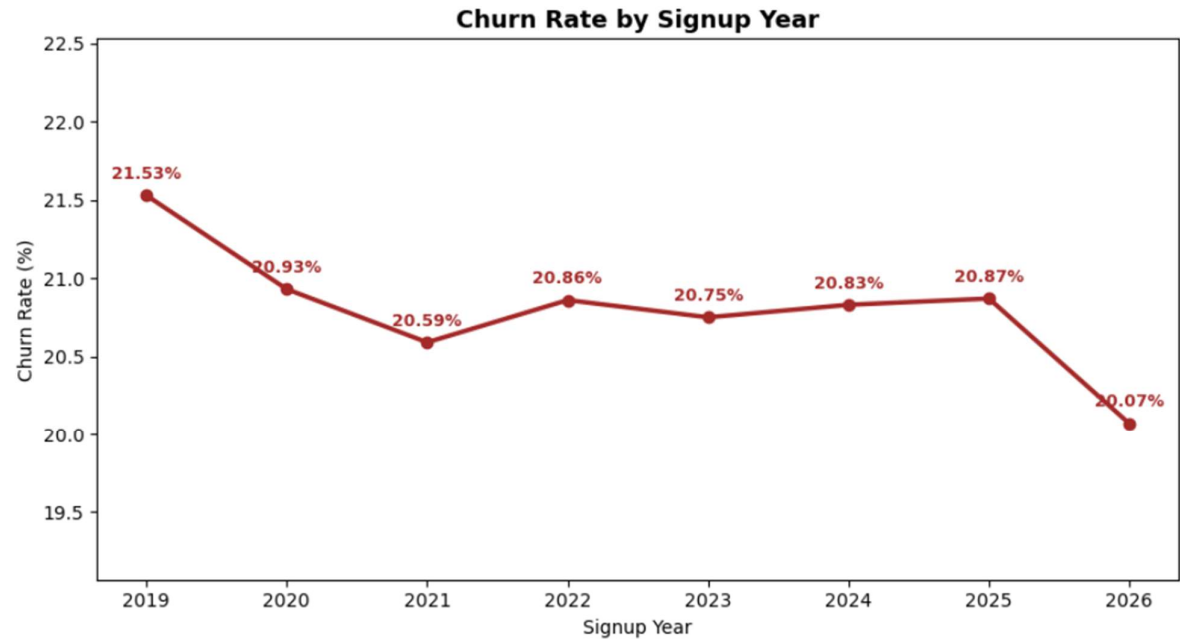
The results are presented in Power BI through **Customer Churn Rate by Support Ticket Volume**.

8.10 Churn by Signup Year

Signup year was analysed to examine whether observed churn rates vary across customer cohorts based on their original signup year.

Signup Year	Customers	Churned	Retained	Churn Rate
2019	480	379	101	78.96%
2020	497	384	113	77.26%
2021	487	374	113	76.80%
2022	491	384	107	78.21%
2023	460	348	112	75.65%
2024	467	356	111	76.23%
2025	349	270	79	77.36%
2026	13	11	2	84.62%

The observed churn rates for the main customer cohorts from **2019 to 2025 range from 75.65% to 78.96%**, indicating relatively limited variation across these signup cohorts. The 2019 cohort records 78.96%, while the 2023 cohort records 75.65%.



The **2026 cohort has an observed churn rate of 84.62%**, but it contains only **13 customers**. Because of this small cohort size, its percentage can change substantially with a small number of customer records and is therefore not directly comparable with the larger historical cohorts.

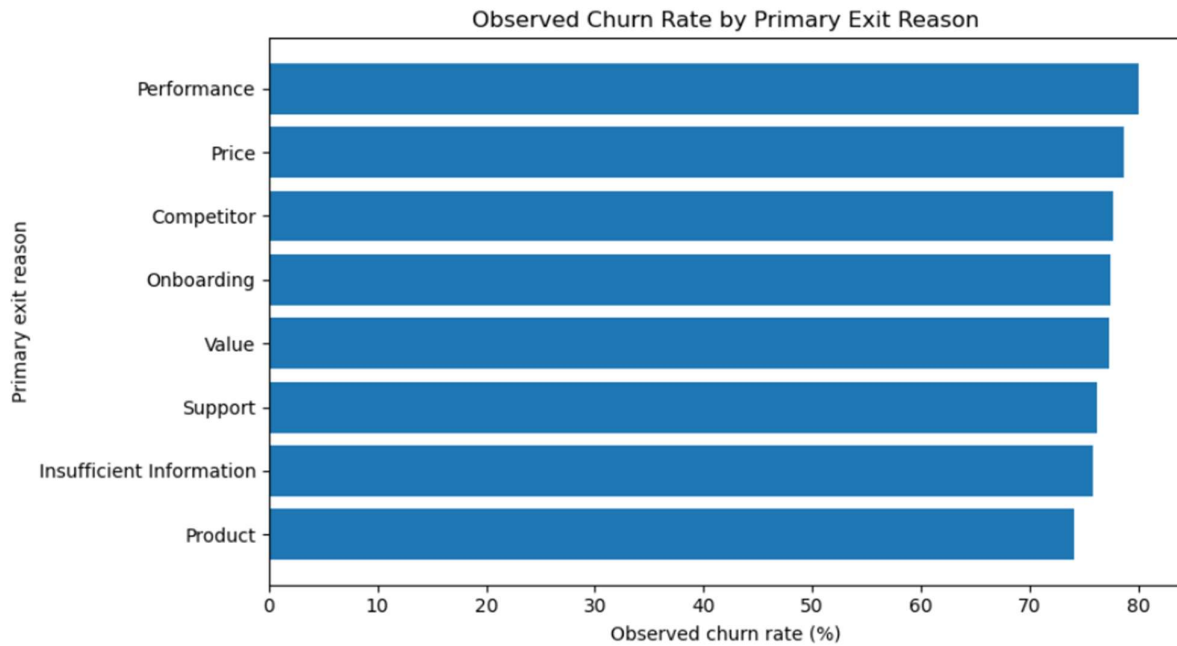
The Power BI analysis presents this relationship through **Customer Cohort Size and Churn Rate by Signup Year**, combining cohort size with observed churn rate.

8.11 Churn by Primary Exit Reason

The integrated analysis was extended to the NLP-derived **Primary Exit Reason** to examine how observed churn varies across the reasons recorded in customer exit interviews. This analysis uses the **3,272 matched interview records**.

Primary Exit Reason	Interview Records	Observed Churn Rate
Performance	342	79.82%
Price	366	78.69%
Competitor	521	77.74%
Onboarding	304	77.63%
Value	808	76.98%
Support	437	76.43%
Insufficient Information	95	75.79%
Product	399	74.19%

The observed churn rates range from **74.19% for Product** to **79.82% for Performance**. Performance, Price, Competitor, and Onboarding are all above the overall interview-level churn rate of **77.20%**, while Value, Support, Insufficient Information, and Product are below it.



The results indicate that different exit reasons are associated with somewhat different observed churn rates. **Performance** has the highest observed churn rate among the classified reasons, while **Product** has the lowest. These differences describe patterns within the matched interview population and do not establish that a particular exit reason caused the customer to churn.

The Power BI page presents this analysis through **Observed Churn Rate by Primary Exit Reason**, alongside the distribution of exit reasons and customer mood.

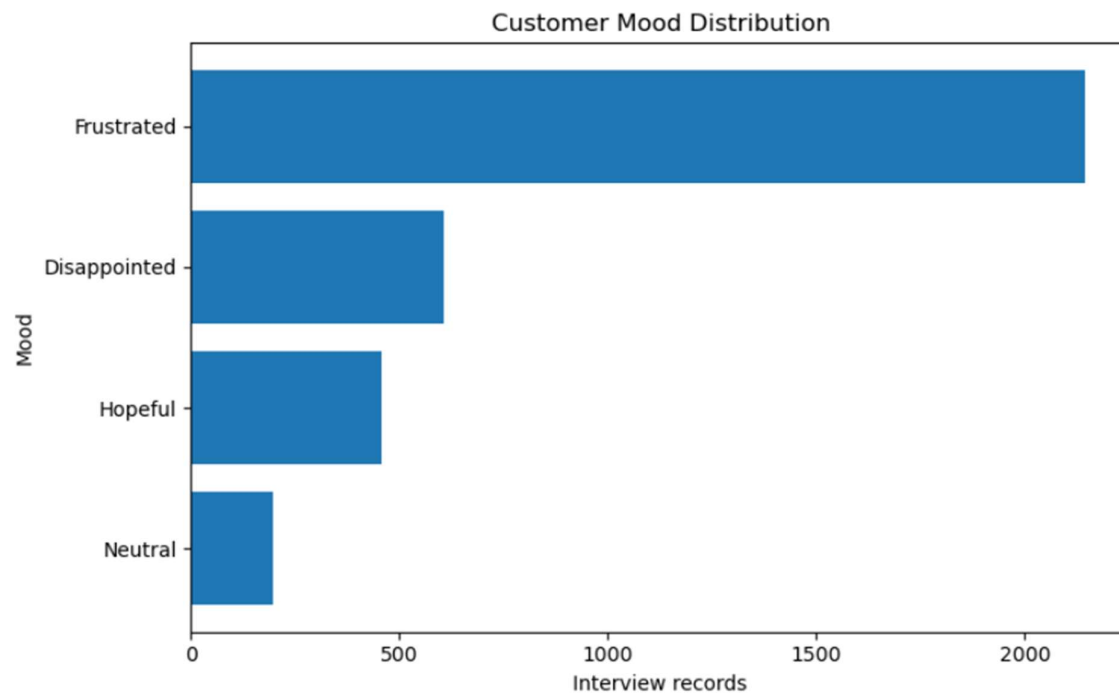
8.12 Customer Mood in Exit Interviews

Customer mood was analysed to understand the emotional tone expressed in matched exit interviews. Four standardized mood categories were used: **Frustrated**, **Disappointed**, **Hopeful**, and **Neutral**.

Customer Mood	Interview Records	Share of Matched Interviews
Frustrated	2,067	63.17%
Disappointed	582	17.79%
Hopeful	435	13.29%
Neutral	188	5.75%
Total	3,272	100.00%

The largest group is **Frustrated**, representing 2,067 matched interviews. Disappointed is the second-largest category with 582 interviews, while Hopeful and Neutral account for 435 and 188 interviews respectively.

Together, **Frustrated and Disappointed interviews account for approximately 81% of the matched interview population**, indicating that negative emotional expressions are prominent within the exit-interview dataset.



Mood provides an additional dimension to the NLP analysis because two customers may report similar business-related exit reasons while expressing different emotional tones. The mood classification is therefore used alongside **Primary Exit Reason**, rather than as a replacement for the underlying reason.

The Power BI analysis presents the distribution through **Customer Mood in Matched Exit Interviews** and further examines how mood composition varies across primary exit reasons.

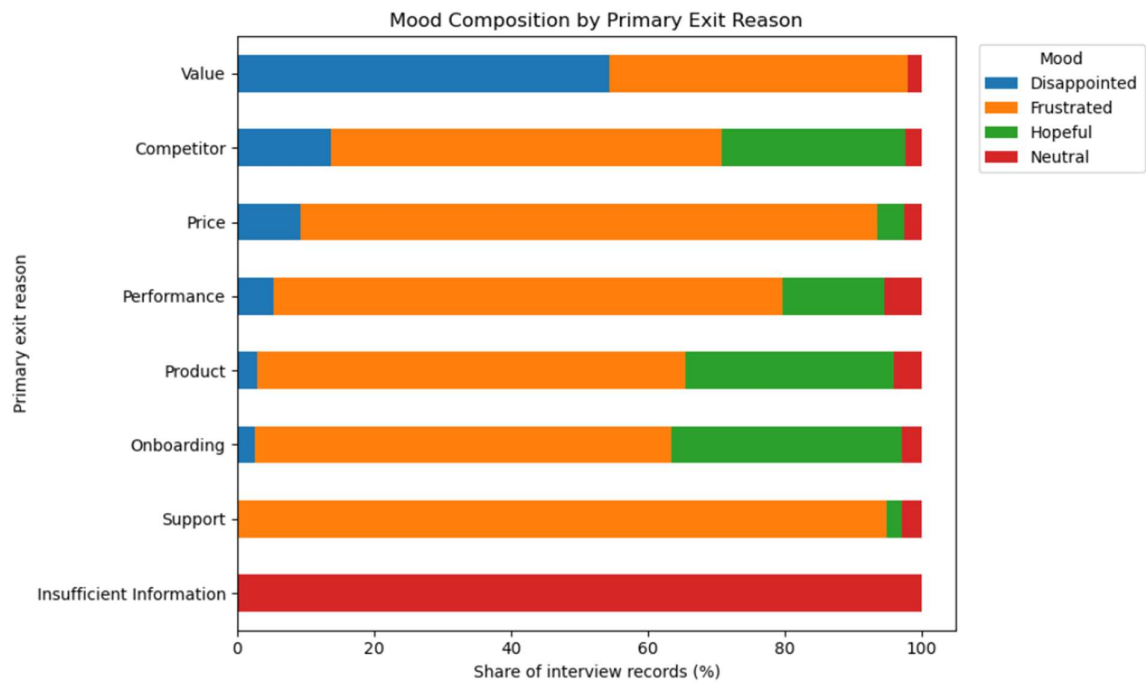
8.13 Mood Composition by Primary Exit Reason

The relationship between **Primary Exit Reason** and **Customer Mood** was examined to understand how emotional tone varies across different reasons identified in the exit-interview text. The analysis uses the full **3,405 NLP-classified interview records**.

Primary Exit Reason	Disappointed	Frustrated	Hopeful	Neutral
Competitor	13.71%	57.04%	26.87%	2.38%
Insufficient Information	0.00%	0.00%	0.00%	100.00%
Onboarding	2.57%	60.77%	33.76%	2.89%
Performance	5.29%	74.37%	14.76%	5.57%
Price	9.16%	84.29%	3.93%	2.62%
Product	2.88%	62.59%	30.46%	4.08%
Support	0.00%	94.90%	2.22%	2.88%
Value	54.35%	43.62%	0.00%	2.03%

The mood composition differs substantially across exit reasons. **Support** is predominantly associated with Frustrated interviews (94.90%), while **Price** also has a high Frustrated share of 84.29%. In contrast, **Value** contains a much larger Disappointed component (54.35%), with 43.62% classified as Frustrated.

Competitor, **Onboarding**, and **Product** show more mixed emotional compositions, with both Frustrated and Hopeful responses represented. **Insufficient Information** contains 95 records, all of which were classified as Neutral because the available interview text did not provide sufficient information to assign a more specific emotional category.



These results add context to the exit-reason analysis: the same broad business category can be expressed with different emotional tones. The analysis therefore considers **reason and mood together** rather than treating exit reason alone as the complete description of the customer's experience.

8.14 Comparative Churn Analysis by Subscription Type and Region

To examine whether observed churn patterns varied when **Subscription Type** and **Region** were considered together, a two-dimensional comparison was created using customer-level churn rates. This analysis extends the individual subscription and regional comparisons by showing the observed churn rate for each subscription-region combination.

Subscription Type	Central	East	North	South	West	Total
Gold	83.33%	100.00%	66.67%	88.89%	100.00%	88.89%
Starter	87.50%	60.00%	86.67%	85.71%	80.00%	81.16%
Standard	77.06%	78.67%	80.71%	79.43%	80.51%	79.43%
Basic	78.26%	76.78%	81.99%	75.88%	77.67%	78.18%
Corporate	100.00%	33.33%	100.00%	75.00%	—	75.00%
Entry	75.00%	87.50%	62.50%	70.00%	76.47%	74.51%
Premium	71.93%	73.17%	69.18%	74.60%	76.07%	72.84%
Pro	50.00%	90.91%	68.42%	66.67%	80.00%	71.64%
Mid	40.00%	81.25%	71.43%	75.00%	62.50%	70.83%
Enterprise	77.78%	75.00%	70.00%	60.00%	66.67%	69.14%
Total	75.68%	77.38%	77.76%	76.51%	78.52%	77.25%

The combined comparison shows that churn rates can vary considerably within the same subscription type across regions. For example, **Standard** ranges from 77.06% in Central to 80.71% in North, while **Enterprise** ranges from 60.00% in South to 77.78% in Central.

Some of the more extreme percentages occur in smaller subscription-region groups. For example, Gold reaches 100% in East and West, while Corporate reaches 100% in Central and North. These percentages are based on relatively small populations, so they should not be treated as directly comparable to the larger Basic or Standard groups.

The Power BI report presents this analysis through the **Observed Churn Rate by Subscription Type and Region** matrix, with the overall customer churn rate of **77.25%** shown as the total reference point.

Key point: The combined view demonstrates that subscription and region can produce different observed churn patterns when analyzed together, but the size of each segment must also be considered when interpreting the percentages.

8.15 Churn Drivers — Decomposition Tree Analysis

The **Decomposition Tree** in Power BI was used to interactively break down the **2,506 churned customers** across multiple customer attributes. The analysis begins with total churned customers and allows the population to be examined through dimensions such as **Subscription Type, Region, Tenure Group, Gender, Age Group, and Spend Group**.

Analysis Component	Role in Decomposition Tree
Churned Customers	Primary measure being analyzed
Subscription Type	Examines churned customers across subscription plans
Region	Examines regional distribution of churned customers
Tenure Group	Examines churned customers by relationship duration
Gender	Examines demographic composition
Age Group	Examines churn across age segments
Spend Group	Examines churned customers across monthly-spend segments

The starting point of the tree is **2,506 churned customers**. At the subscription level, the largest churned-customer populations are **Basic (903), Standard (869), and Premium (456)**, followed by Enterprise (56), Starter (56), Pro (48), Gold (40), Entry (38), Mid (34), and Corporate (6).

The tree can then be expanded interactively to examine combinations of these attributes. For example, a user can first select a subscription type and then break that population down by region, tenure, gender, age, or spending group. This makes the visual useful for exploring **multi-dimensional customer segments** rather than examining each characteristic separately.

The decomposition tree is therefore used as an **exploratory analysis tool**. It identifies how the observed churned-customer population is distributed across combinations of customer attributes; it does not establish that any particular attribute causes churn.

8.16 Key Findings from Churn Analysis

The churn analysis combined customer-level segmentation, comparative analysis, and exit-interview NLP results to identify the main observed patterns in the integrated dataset.

Analysis Area	Key Finding
Overall Churn	2,506 of 3,244 customers were churned , giving an observed customer churn rate of 77.25% .
Subscription Type	Observed churn rates ranged from 69.14% for Enterprise to 88.89% for Gold . Small subscription groups require additional attention when interpreting percentages.

<i>Region</i>	Regional churn rates were relatively close, ranging from 75.68% in Central to 78.52% in West .
<i>Gender</i>	Churn rates varied across gender groups, although some groups contained relatively few customers.
<i>Age</i>	The 35–44 group recorded 79.50% churn, while the age pattern was not a simple increasing or decreasing relationship.
<i>Tenure</i>	Churn varied across tenure groups without a consistent linear pattern.
<i>Monthly Spend</i>	Churn also varied across spending groups without a simple monotonic relationship.
<i>Support Tickets</i>	Observed churn increased from 74.82% for customers with 0 tickets to 82.27% for customers with 4+ tickets .
<i>Exit Reason</i>	Interview-level observed churn ranged from 74.19% for Product to 79.82% for Performance .
<i>Customer Mood</i>	Among matched interviews, Frustrated was the largest mood category, followed by Disappointed , Hopeful , and Neutral .
<i>Reason × Mood</i>	Support and Price were predominantly classified as Frustrated, while Value contained a larger Disappointed component.
<i>Comparative Segmentation</i>	Subscription-region combinations showed different observed churn rates, demonstrating that segment patterns can change when multiple attributes are considered together.
<i>Decomposition Analysis</i>	The Power BI Decomposition Tree provided an interactive way to examine the 2,506 churned customers across subscription, region, tenure, gender, age, and spending dimensions.

Overall, the analysis shows that churn is **not explained by a single customer characteristic**. Different dimensions show different levels of variation, while the integrated NLP analysis adds information about **why customers reported leaving and the emotional tone associated with those interviews**.

The findings represent **observed associations and segment patterns**, rather than causal relationships. Small customer groups also require particular care because their percentages can change substantially with a small number of records.

9. Power BI Dashboard and Reporting

The validated customer-level and interview-level analysis was translated into a five-page Power BI report. The dashboard was designed to move from overall churn measurement to customer segmentation, exit-interview analysis, and interactive exploration of churn drivers.

Dashboard Page	Purpose	Main Visuals / Measures
Page 1 — AI-Powered Customer Churn Analytics	Provides the overall SaaS customer retention and churn overview	Total Customers, Churned Customers, Retained Customers, Customer Churn Rate, Interview Records; churn by Subscription Type, Region, Support Tickets, and Tenure
Page 2 — Customer Demographics & Churn	Examines churn across customer demographic and behavioral characteristics	Age, Gender, Monthly Spend, and Signup Year
Page 3 — Exit Interview & NLP Analysis	Connects customer exit feedback with NLP-derived reasons and mood	Primary Exit Reasons, Customer Mood, Mood Composition by Exit Reason, Observed Churn Rate by Exit Reason
Page 4 — Churn Drivers & Advanced Analysis	Enables interactive exploration of the churned-customer population	Decomposition Tree across Subscription Type, Region, Tenure, Gender, Age, Spend, and other available attributes
Page 5 — Churn Drivers — Comparative Analysis	Examines combined segment patterns	Subscription Type × Region churn-rate matrix and churned-customer composition

Dashboard Design and Analytical Logic

The dashboard uses **Customer Level** as the primary customer-level analytical table, containing **3,244 unique customers**, while the integrated interview table contains **3,272 interview records**. This distinction allows customer-level churn metrics to represent each customer once while preserving individual interview records for NLP analysis.

The main dashboard KPIs report **3,244 total customers, 2,506 churned customers, 738 retained customers, and a 77.25% customer churn rate**. The dashboard also reports 3,272 matched interview records.

The comparative pages extend the analysis beyond individual variables. For example, the Subscription Type × Region matrix shows that observed churn can differ within the same subscription category depending on region, while the 100% stacked composition chart shows the distribution of churned customers across subscription types within each region.

The NLP page provides a separate analytical layer. It shows the distribution of the eight primary exit-reason categories, customer mood across matched interviews, mood composition by exit reason, and observed churn rates associated with the classified exit reasons.

The dashboard therefore functions as the final analytical interface of the project, bringing together **data quality preparation, customer segmentation, churn analysis, NLP classification, and comparative exploration** in a single reporting environment.

10. Recommendations

The analysis provides several areas where the observed churn patterns and exit-interview information can be translated into practical customer-retention actions. The recommendations below are based on the patterns identified in the project rather than treating any individual variable as a proven cause of churn.

Recommendation Area	Evidence from Analysis	Recommended Action
Subscription Management	Churn rates differed across subscription types, from 69.14% to 88.89%.	Monitor churn by subscription type and investigate the customer experience, pricing, and usage patterns within individual plans.
Support Experience	Customers with 4+ support tickets had an observed churn rate of 82.27%, compared with 74.82% for customers with no tickets.	Establish follow-up workflows for customers with repeated support interactions and review recurring support issues.
Exit-Reason Monitoring	Performance, Price, Competitor, and Onboarding were among the exit-reason categories with relatively high observed churn rates.	Track exit reasons regularly and connect them with product, pricing, onboarding, and competitive-response initiatives.
Customer Feedback and Mood	Frustrated and Disappointed were the largest mood categories in matched exit interviews.	Use NLP-based feedback monitoring to identify recurring negative customer experiences and prioritize areas for service improvement.
Segment-Level Monitoring	Subscription × Region analysis showed that churn patterns can vary across combinations of customer attributes.	Use multidimensional dashboard analysis rather than relying only on overall churn or a single customer attribute.
Early-Warning Analytics	The current project identifies observed associations but does not generate individual future churn probabilities.	A future predictive model could use validated customer and NLP features to estimate churn probability and support proactive retention workflows.
Dashboard-Based Monitoring	The five-page Power BI report consolidates customer, churn, NLP, and comparative analysis.	Use the dashboard as a recurring monitoring framework and refresh the underlying analytical data as new customer and interview records become available.

These recommendations should be treated as **operational applications of the observed patterns**, not as evidence that any single factor directly causes churn. The current project is primarily descriptive and exploratory; predictive modeling would be a logical next development for estimating future customer-level churn.

11. Project Limitations

The project provides an integrated view of customer churn using structured customer information and exit-interview text. However, several limitations define how the results should be used and interpreted.

Limitation	Impact on Analysis
Historical Dataset	The analysis reflects the customer and interview data available in the project and may not represent future customer behaviour.
Descriptive and Diagnostic Focus	The project identifies observed churn patterns and associations but does not build a predictive model that estimates future individual customer churn probability.
Limited External Factors	Factors such as competitor actions, broader market conditions, customer-specific business circumstances, and other external influences were not available in the dataset.
Exit-Interview Coverage	Only 3,272 interview records could be successfully matched to the customer master. The remaining 133 NLP records were retained separately because no corresponding customer record was available.
Interview Text Quality	Some interview records contained limited textual information. These records were classified as Insufficient Information where the available text did not support a more specific exit-reason classification.
Small Segment Sizes	Some subscription, demographic, and subscription-region groups contain relatively few customers. Their percentages can therefore change substantially with a small number of records.
Date Interpretation	Some raw signup dates were ambiguous between DD-MM-YYYY and MM-DD-YYYY formats. The ambiguous records were standardized using the dominant dataset convention, with the original raw date retained for auditability.
Imputation	Missing analytical customer attributes were addressed using documented peer-based imputation. The original values and imputation indicators were retained, but imputed values remain estimates rather than directly observed values.
Causal Inference	The analysis identifies associations between customer characteristics, exit reasons, mood, and observed churn. It does not establish that any particular characteristic or exit reason directly caused churn.
Current Dashboard Scope	Power BI provides interactive descriptive and comparative analysis, but it does not currently provide real-time churn prediction or automated retention decisions.

These limitations do not prevent the dashboard from being used for **descriptive customer analysis and decision support**. Instead, they define the appropriate interpretation of the results and provide a clear foundation for future development. The project documentation also identifies predictive churn modelling, real-time monitoring, additional behavioural data, and customer lifetime value analysis as potential extensions.

12. Future Scope

The current project provides a descriptive and diagnostic foundation for SaaS customer churn analysis by combining structured customer information, exit-interview NLP classification, customer-level churn analysis, and Power BI reporting. The validated analytical dataset can be extended into more advanced retention and customer-intelligence applications.

Future Development	Proposed Extension	Expected Analytical Value
Predictive Churn Modeling	Develop supervised machine-learning models using customer attributes, support activity, tenure, spending, subscription information, and NLP-derived features.	Estimate the probability that an individual customer may churn in a future period.
Advanced NLP Analysis	Extend the current exit-reason and mood classification using embeddings, topic modeling, aspect-based sentiment, and more detailed emotion analysis.	Extract deeper themes and identify specific product, pricing, support, or performance issues from customer feedback.
Customer Behaviour Data	Integrate product usage, login frequency, feature adoption, session activity, plan utilization, and engagement metrics.	Add behavioural signals that are not available in the current customer dataset.
Customer Lifetime Value Analysis	Combine churn behaviour with monthly spending, tenure, and future-value estimates.	Understand churn in the context of customer value and support retention prioritization.
Real-Time / Automated Monitoring	Connect Power BI with regularly refreshed customer and support data and introduce automated monitoring or alerts.	Enable earlier identification of changing churn patterns rather than relying only on periodic analysis.
Retention Intervention Tracking	Record retention actions such as offers, support escalation, onboarding assistance, or product outreach and evaluate subsequent customer outcomes.	Measure whether specific retention actions are associated with improved customer retention.
Model Monitoring and Retraining	If predictive models are introduced, monitor model performance, data drift, feature changes, and classification quality over time.	Help maintain reliability as customer behaviour and SaaS business conditions change.
Expanded Dashboard Analytics	Extend the Power BI report with predictive-risk indicators, customer-value segments, behavioural trends, and automated alerts.	Move the dashboard from descriptive reporting toward operational customer-retention monitoring.

The progression can therefore move from the current **descriptive and diagnostic analysis** toward **predictive and eventually prescriptive customer-retention analytics**. The existing data-cleaning, validation, NLP integration, customer-level analysis, and Power BI framework provides the foundation for these extensions while preserving the project's emphasis on data quality and traceability.

13. Conclusion

This project developed an integrated **AI-powered customer churn analytics framework for the SaaS industry**, combining structured customer data with exit-interview NLP. After systematic cleaning and validation, the customer dataset was reduced from **95,380 raw records to 95,000 unique customers**. The integrated interview analysis covered **3,272 matched interview records representing 3,244 unique customers**, with **2,506 churned customers and 738 retained customers**, resulting in an observed customer churn rate of **77.25%**.

The analysis identified observed churn patterns across subscription type, region, demographics, tenure, spending, support activity, exit reasons, and customer mood. The final **five-page Power BI dashboard** brought these findings together through interactive churn KPIs, customer segmentation, NLP insights, decomposition analysis, and comparative analysis. The project provides a validated foundation for future predictive churn modeling, advanced NLP, behavioural analysis, and automated customer-retention monitoring.

Appendices

APPENDIX	TITLE	CONTENTS
APPENDIX A	Data Dictionary	Customer and exit-interview fields, definitions, and analytical roles
APPENDIX B	Data Cleaning & Validation Evidence	Duplicate analysis, conflicting IDs, missing values, numerical validation, date validation, and final checks
APPENDIX C	NLP Classification Framework	Exit-reason categories, mood categories, classification structure, and examples
APPENDIX D	Customer–Interview Integration	Matching results, unmatched interviews, multiple-interview records, and integration summary
APPENDIX E	Imputation Audit	Detailed record of the 54 imputed analytical values across 44 records
APPENDIX F	Detailed Churn Analysis Tables	Full customer-level churn tables and supporting analytical results
APPENDIX G	Detailed NLP Analysis Tables	Exit-reason, mood, reason × mood, reason × churn, and related results
APPENDIX H	Additional Notebook Analytical Visualizations	Important notebook charts not already presented in the main report
APPENDIX I	Technical Workflow & Tools	Python, SQL/BigQuery, Excel, NLP, and Power BI workflow
APPENDIX J	Selected Python / SQL Code	Key code used for cleaning, validation, integration, analysis, and preparation